

Section 9.1 Worksheet: Systems of Linear Inequalities

Instructions. Show boundary equations, test-point calculations, intersection systems, and interpretations.

1. Determine whether each point satisfies $2x - y > 1$: $(2, 1)$, $(0, 0)$, and $(-1, -4)$.
2. Graph $3x + 2y \leq 12$. State whether the boundary is solid or dashed, find both intercepts, and identify the shaded side using a test point.

3. For

$$x \geq 0, \quad y \geq 0, \quad x + y \leq 6, \quad 2x + y \leq 8,$$

find every vertex, list the active constraints at $(2, 4)$, and determine whether the region is bounded.

4. Determine whether the region $x \geq 0$, $y \geq 0$, and $x - y \geq 1$ is bounded. Give one feasible point and one direction in which feasible points continue indefinitely.

5. A fictional travel week includes x driving excursions and y hiking excursions. There are at most seven excursions; driving uses three fuel units and hiking uses one; at most fifteen fuel units are available; and at least two hiking excursions are required. Write the complete system, including nonnegativity.
6. Prove that the intersection of any collection of linear half-planes is convex. Your proof must begin with two feasible points and an arbitrary convex combination.
7. Explain why testing a single point works after a boundary line is drawn. State the role of continuity or the impossibility of changing sign without crossing the boundary.

Section 9.2 Worksheet: Graphical Linear Programming

Part A. Maximize $Z = 50x + 40y$ subject to

$$x, y \geq 0, \quad x + y \leq 10, \quad 2x + y \leq 16, \quad x + 3y \leq 21.$$

1. Find every feasible vertex. Show the boundary intersections used and reject any infeasible intersections.
2. Evaluate Z at every vertex and state the maximum.
3. At the optimizer, identify all active constraints and calculate every slack.
4. Replace the objective by $Q = 2x + y$. Determine the full set of optimal solutions and explain why it is not a single point.

Part B. Applications and theory

5. A fictional R&B and Backstreet Boys listening event offers x premium sessions and y standard sessions. The limits are

$$x + y \leq 9, \quad 2x + y \leq 14, \quad x + 2y \leq 16, \quad x, y \geq 0.$$

Revenue is $R = 80x + 60y$. Find the optimal plan and revenue by the vertex method.

6. Determine whether maximizing $x + y$ subject to $x - y \geq 1$, $x, y \geq 0$ has a finite solution. Justify geometrically and algebraically.
7. Prove the vertex principle for a polygonal feasible region using convex combinations and linearity of the objective.

Section 9.3 Worksheet: Simplex Maximization

Use exact fractions throughout.

Part A. Complete simplex solution

$$\max z = 3x_1 + 2x_2$$

subject to

$$x_1 \leq 4, \quad x_2 \leq 5, \quad x_1 + x_2 \leq 6, \quad x_1, x_2 \geq 0.$$

1. Introduce slack variables, write the equation system, and construct the initial tableau.
2. For every pivot, identify the entering variable, calculate all eligible ratios, identify the departing variable, and state the pivot.
3. Show every row operation and every resulting tableau.

4. Report the optimal decision variables, objective value, and all slack values. Verify the original constraints.
5. Rewrite the final objective row as an inequality proving that no feasible solution has a larger objective value. **Part B. Reading and diagnosing tableaus**
6. In a feasible maximization tableau, a nonbasic variable has reduced cost 0. What does this suggest about alternate optimal solutions?
7. An entering column has no positive entry in the constraint rows. Prove that the objective is unbounded in the entering direction.
8. Explain degeneracy and why a zero ratio can produce a pivot with no objective improvement. State one anti-cycling rule.

Section 9.4 Worksheet: Minimization and Duality

Consider

$$\min w = 5x_1 + 4x_2$$

subject to

$$x_1 + x_2 \geq 5, \quad 2x_1 + x_2 \geq 8, \quad x_1, x_2 \geq 0.$$

1. Construct the dual maximization problem. Explain how each primal row, column, and coefficient is used.
2. Solve the dual by simplex. Show every tableau, ratio, pivot, and row operation.
3. Use strong duality to state the primal minimum. Recover and verify the primal optimizer.
4. Check every complementary-slackness product.

5. Prove weak duality for the orientation $A\mathbf{x} \geq \mathbf{b}$, $A^T\mathbf{y} \leq \mathbf{c}$.
6. Explain why equality of one primal feasible value and one dual feasible value proves both solutions optimal.
7. Construct, but do not solve, the dual of
- $$\min 7x_1 + 5x_2$$
- subject to $x_1 + x_2 \geq 6$, $3x_1 + x_2 \geq 10$, and $x_1, x_2 \geq 0$.
8. Interpret the dual variables as shadow prices. State why this interpretation is local rather than valid for arbitrarily large right-hand-side changes.

Section 9.5 Worksheet: Mixed Constraints

Part A. Two-phase simplex

$$\max z = 4x_1 + 3x_2$$

subject to

$$x_1 + x_2 \leq 5, \quad 2x_1 + x_2 \geq 4, \quad x_1, x_2 \geq 0.$$

1. Introduce the slack, surplus, and artificial variables and write the equality system.
2. Form the canonical Phase I tableau. Explain why the artificial variable must be eliminated from the Phase I objective row before pivoting.
3. Carry out Phase I with every ratio and row operation. Prove feasibility from the resulting Phase I objective.
4. Remove the artificial variable, restore the original objective, and complete Phase II with every tableau.

5. Report the optimizer, maximum, slack, and surplus. Verify the original inequalities. **Part B. Structure**

and modeling

6. For each constraint, state the required added variables:

$$2x_1 + x_2 \leq 8, \quad x_1 + 3x_2 \geq 6, \quad x_1 - x_2 = 2.$$

7. Prove that the original constraints are feasible if and only if the Phase I minimum is zero.

8. Convert the mixed minimization objective $w = 7x_1 + 5x_2$ into a maximization objective. Explain what happens to the constraints.

9. Two depots have capacities 400 and 300 units. Two hiking centers require at least 250 and 300 units. Let x_{ij} be the shipment from depot i to center j . Formulate all supply, demand, and nonnegativity constraints. If unit costs are

$$\begin{bmatrix} 4 & 6 \\ 5 & 3 \end{bmatrix},$$

write the cost objective.